

Distances in discrete space-time

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Background: Motivation I

- ▶ Modern Physics has a problem
- ▶ The Standard Model fails to account for gravity which is instead characterised by General Relativity
- ▶ One resolution to this incompatibility is Causal Set Theory

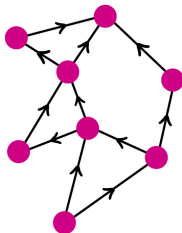
Causal Set Theory $\sim$ Standard Model $\cup$ General Relativity

Hopefully!

Background: Motivation II

- ▶ In the same way continuous matter is discretised into atoms, **space-time is discretised** into space-time 'atoms'
- ▶ These 'atoms' constitute the elements of a set and we place a partial ordering on them to impose causality

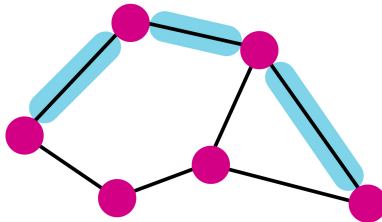
$$S = \{x_1 \preceq x_2 \preceq x_3 \cdots\} \iff$$



- ▶ We can map this set to a Directed Acyclic Graph (DAG)

Background: Graphs I

- ▶ A graph is constructed from a set of nodes and a set of edges which connect those nodes



- ▶ A path describes a sequence of unique nodes and the edges connecting them
- ▶ The highlighted edges draw a possible path across the graph

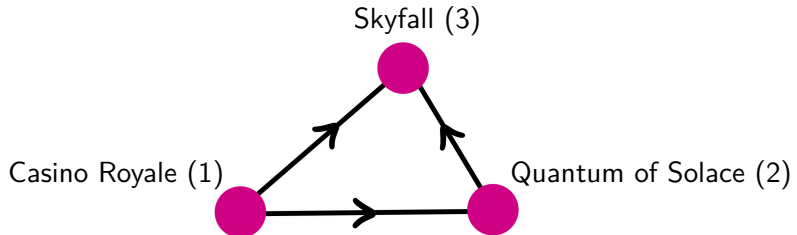
Background: Graphs II

- ▶ A DAG is a class of graph with special properties

DAG Properties

1. *Directed*: Edges are directed, acting as '**one way streets**'.
Causal links go in one direction from what causes to what is caused
 2. *Acyclic*: Paths that start and end at the same node are not allowed. An event cannot cause itself
- ▶ These properties ensure the points obey causality by imposing an 'arrow of time'

Background: Graphs III



- ▶ There is a causal link from Casino Royale to Skyfall but not one the other way around
- ▶ Skyfall requires both Casino Royale and Quantum of Solace but Quantum of Solace only requires Casino Royale
- ▶ Casino Royale cannot cycle back to itself because it cannot cause itself

Paths: Distance I

- ▶ Space-time 'atoms' have no notion of position such that the distance between them cannot depend on position
- ▶ We define the path based distance between two points as the number of edges along a path linking them

Myrheim Length Conjecture

In the continuum limit, $\rho \rightarrow \infty$, the **longest** path in the DAG corresponds to the geodesic (the trajectory which minimises the action in continuous space-time)[1]

- ▶ This longest path maximises the path based distance
- ▶ Agreement with this conjecture is established through simulation by us (for testing) and by others[2]

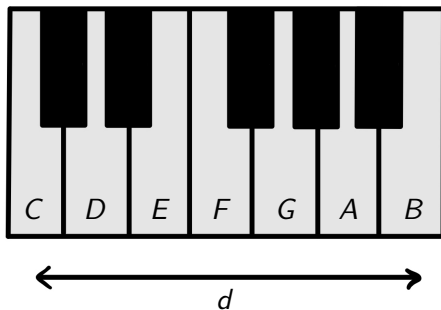
Paths: Distance II

- ▶ We however observe space-time as a Lorentzian manifold (locally Minkowski) such that position based distance must be an emergent property of the causal set
- ▶ We can build a toy model to investigate the correspondence between position based paths and the geodesic by introducing the Minkowski distance

$$\ell(A, B) = \left(\sum_{i=1}^n |A_i - B_i|^p \right)^{\frac{1}{p}}$$

- ▶ We sum over the co-ordinates of A and B
- ▶ Deviations from $p = 2$ (Euclidean) can change the triangle inequality

Paths: Distance III



- ▶ Path based: $d = 6$
- ▶ Position based: $d = \ell(C, D) + \ell(D, E) + \cdots + \ell(A, B)$

Paths: Algorithms I

- ▶ We require a systematic method of finding paths

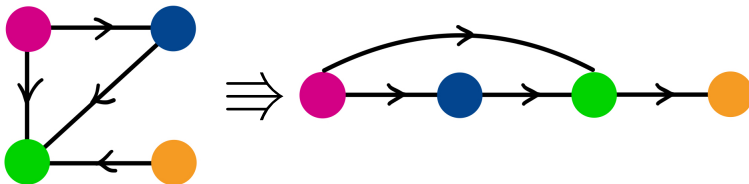
Dijkstra's Shortest Path Algorithm

1. Set distance to source $d[0] = 0$
2. Set distance to neighbours $d[n+1] = \min\{d[n] + \ell(n, n+1)\}$
3. Repeat (2) for unvisited neighbours
4. Stop when the sink node is reached

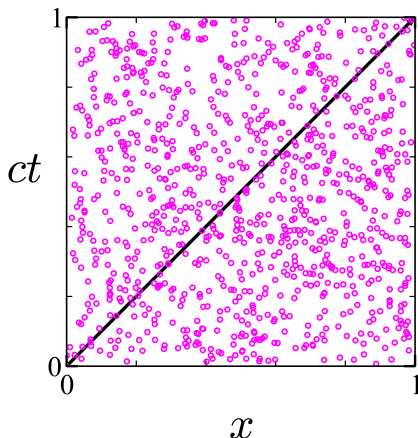
- ▶ Swapping min for max, the algorithm finds the longest path
- ▶ It is instructive to study both in this context

Paths: Algorithms II

- ▶ We topologically sort our graph before finding the path to achieve a time complexity of $\mathcal{O}(N + E)[3]$



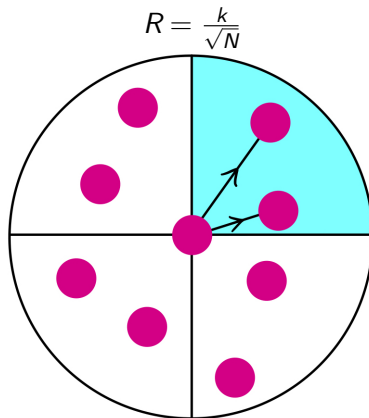
Simulation: Node Generation



- ▶ We generate finite DAGs to represent a window of $1+1$ space-time
- ▶ We sprinkle $N - 2$ points with **independently random co-ordinates** such that they are Poisson distributed between the source $(0,0)$ and sink $(1,1)$
- ▶ The geodesic is a straight line between the source and sink

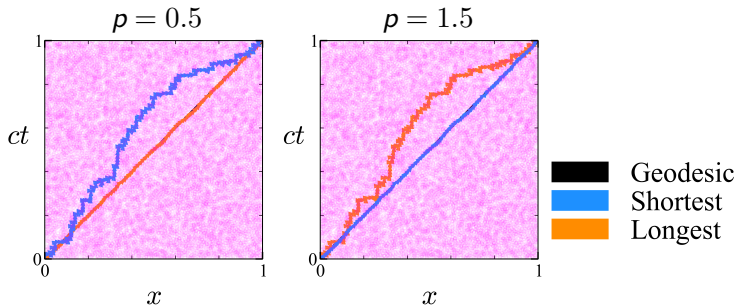
Simulation: Edge Generation

- ▶ We generate outgoing edges to points in the **future light cone** with a cutoff, R
- ▶ k roughly counts how many generations of neighbours we allow interactions with
- ▶ This ensures the DAG conforms to the arrow of time and eliminates long distance interactions



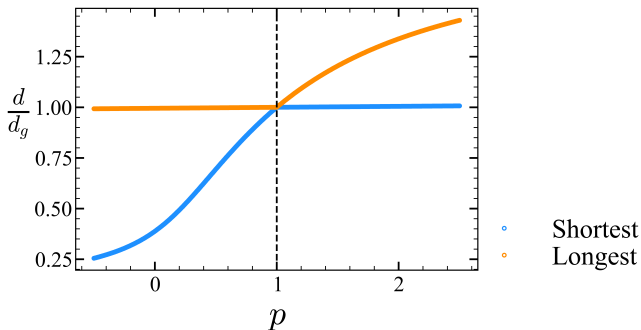
Results: Position Based Distance

- ▶ We see the longest and shortest paths for $p = 0.5$ and $p = 1.5$ for $N = 30000$ nodes and compare to the geodesic
- ▶ The path that well approximates the geodesic inverts
- ▶ The inversion is not perfectly symmetric as seen by minor differences between the paths



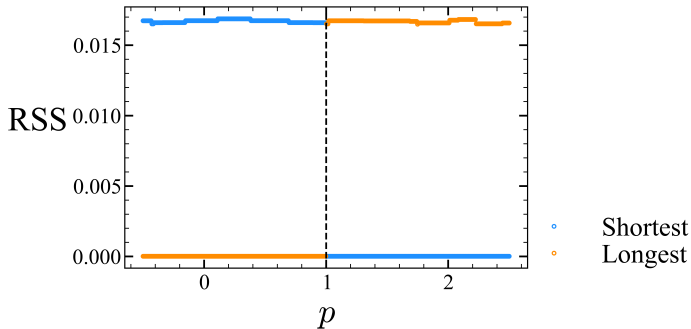
Results: Measure of Agreement I

- ▶ We look at path distance for 500 values, $-0.5 \leq p \leq 2.5$
- ▶ Each curve sees a region where it matches geodesic and a region where there is continuous growth
- ▶ The longest path matches the geodesic for $p < 1$, then diverges with the shortest path taking its place for $p > 1$



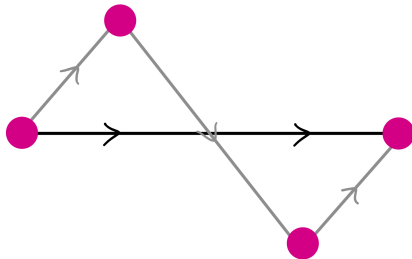
Results: Measure of Agreement II

- ▶ We look at the Residual Sum of Squares (RSS) for identical p
- ▶ The asymmetric inversion is represented by the finite step
- ▶ We expect inversion as crossing $p = 1$ alters the definition of short to what we traditionally think of as long



Results: Discussion I

- ▶ One explanation for the paradoxical equivalence between the longest path and the geodesic (shortest path) is jaggedness
- ▶ The longest path is the most jagged and the most jagged follows the geodesic



- ▶ Further investigation would look at quantifying this jaggedness and at the stability of the RSS

Results: Discussion II

Alas!

- ▶ This is a toy model and it is hard to discuss these plots and their significance in the context of causal set theory without significant conjecture
- ▶ The analysis we perform regards paths in DAGs using the Minkowski distance such that it is very general and has broad applications outside of physics, in data science and elsewhere

Conclusion

- ▶ The Myrheim length conjecture states that the longest path by path based distance corresponds to the geodesic
- ▶ We build on this by investigating position based distance using the Minkowski distance
- ▶ We find that for $p < 1$, the longest path well approximates the geodesic
- ▶ For $p > 1$, we see a path inversion where it is instead the shortest path the well approximates the geodesic
- ▶ The inversion is asymmetric

Acknowledgements

- ▶ I would like to thank my partner Daniel Gaivão Lozano, without whom I would have had a very boring time
- ▶ I would also like to thank my supervisor Dr. Tim Evans

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